

PLACES AND FLOWS CHALLENGE

Places and Flows Contacts:

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1. Overview

This challenge invites you to explore how machine learning can predict visitor emotions at historical fort sites based on environmental features. You will design, implement, and test a predictive system using spatial data (land cover, elevation, terrain data) combined with physiological measurements (skin conductance as a proxy for emotional arousal).

Understanding visitor emotions at specific locations is crucial for heritage site management and visitor experience design, especially *before* managers start to intervene with new buildings, bulldozers, etc. Traditional approaches rely on questionnaires or manual observation, which are time-intensive and limited in spatial resolution. Your task is to develop a machine learning model that can predict emotional arousal based on environmental characteristics.

Specifically, the models developed in this challenge should be compatible with future interactive landscape mapping systems. In such systems, users can change landscape features (e.g., elevation, path width, or vegetation), after which the spatial features are recomputed and the model immediately updates a map showing predicted emotional arousal. Although this challenge uses pre-recorded physiological data, it provides the predictive foundation needed for interactive and tangible landscape applications.

2. The Challenge

Goal: Create a machine learning model to predict emotional arousal at fort locations based on land cover, elevation, and terrain characteristics.

Your task: Develop a predictive model that takes spatial/environmental features as input and predicts skin conductance responses (emotional arousal) as output. Your solution should:

- Combine the two datasets: Location+SkinConductance data (what visitors felt where) with Landcover+Elevation+Geomorphon data (environmental characteristics of those locations).
- Develop meaningful contextual variables (e.g., time, speed, position in sequence) of location + skin conductance data
- Develop meaningful contextual variables (e.g., point, buffer, viewshed) from the spatial data that might influence emotional responses.
- Train and validate a machine learning model that predicts emotional arousal.
- Design the model and feature pipeline in a way that supports rapid recalculation of predicted emotional arousal when environmental features are modified, as required for interactive landscape mapping.
- Evaluate model performance using appropriate metrics.

3. The Data

The data provided by Place and Flows for this project are available at: https://edubuas-my.sharepoint.com/:f:/g/personal/soaresdelima_e_buas_nl/IgAxUtrTxyx-SrV4GPsnrn-uAetBSI2vgfxYRPJ2tcdaK14?e=hpU0BV

The data is organized in two separate folders:

- **Folder 1: Location+SkinConductance** - contains physiological and GPS data from visitors across four forts:
 - Belgium: Fort Leopold (Diest), Fort van Lier
 - Netherlands: Fort Sabina, Fort de Roovere
- **Folder 2: Landcover+Elevation+Geomorphon** - contains environmental characteristics for each fort area in GeoJSON format.

See the README files in each folder for detailed column descriptions.

4. Technical Considerations

This is an open problem with unique challenges:

- Spatial data handling: GPS points need to be matched with environmental polygons.
- Temporal autocorrelation: consecutive measurements from the same person are highly correlated; different people have different baseline arousal levels.
- Imbalanced spatial coverage: some areas have more measurements than others.

- High-frequency data: 64 measurements per second means millions of data points.

We encourage you to be creative in addressing these challenges. Explore different solutions, experiment with various approaches, and justify your decisions based on what you learn from the data.

5. Communication Related to this Challenge

A dedicated Teams group chat (“Places and Flows Challenge”) has been set up for this challenge. If you have any questions, encounter issues, or wish to discuss any aspect of the challenge, please use this chat.

You are encouraged to use the group chat to share your results and exchange ideas. Discussing concepts and sharing materials and tips that may benefit others is highly encouraged.

6. Timeline

1. **Kick-off Presentation:** Monday, January 19, 2026, 11:00 AM - 11:30 AM
 - Format: Online (Teams invitation will be sent)
 - Content: Detailed challenge information, research context, Q&A
 - Attendance: MANDATORY
2. **Work Period:** January 19 to January 30, 2026
 - You are allowed to work on the challenge remotely or on campus (the DataLab rooms will be available).
 - Places and Flows Lab can also offer two workstations to students who are interested in working there (please contact Ondrej and Sait to confirm availability).
3. **Final Presentation:** Friday, January 30, 2026, 2:00 PM - 3:30 PM
 - Format: Online (Teams invitation will be sent)
 - Content: Present your model, approach, and results. More information about format and time limit will be shared before the session.
 - Attendance: MANDATORY

7. Evidencing the Results of the Challenge

The results of the challenge must be submitted on the Brightspace assignment created for the challenge:

Brightspace → General information ADSAI 2025-2026 → Course Elements → Assignments → Places and Flows

Required Deliverables:

- 1. Working code** (Python scripts or Jupyter notebooks)
- 2. Documentation explaining:**
 - Your approach and methodology
 - Model performance analysis
 - Discussion of findings

Submission Deadline:

- Friday, January 30, 2026, before 23:59

The documentation elements can be reported in a separated PDF document or in the same Jupyter notebook where you developed and evaluated your model.

Please note that presenting your results in the final presentation is MANDATORY. Only submitting the deliverables via Brightspace will not be accepted.